# Appendix: Theoretical Analysis of the Augmented Synthetic Historical Control Estimator (ASHC)

## Setup and Assumptions

For the purposes of self-containment, let $\mathcal{T} = \{1, 2, \dots, T\}$ index time at a monthly frequency. Define the pre-treatment period as $\mathcal{T}_1 := \{t \in \mathcal{T} : t \leq T_0\}$ and the post-treatment period as $\mathcal{T}_2 := \{t \in \mathcal{T} : t > T_0\}$. The number of post-treatment periods is $n = T - T_0$. Let $m \in \mathbb{N}$ denote the evaluation window length. Define the evaluation period as the final $m$ months of the pre-treatment period $\mathcal{T}_{\text{eval}} := \{T_0 - m + 1, \dots, T_0\} \subset \mathcal{T}_1$.

Let $\mathbf{y} = (y_1, y_2, \dots, y_T)^\top \in \mathbb{R}^T$ denote the observed outcome vector for the treated unit. Define the pre-treatment outcome vector $\mathbf{y}_{\text{pre}} := \mathbf{y}_{\mathcal{T}_1} \in \mathbb{R}^{T_0}$, the evaluation subvector $\mathbf{y}_{\text{eval}} := \mathbf{y}_{\mathcal{T}_{\text{eval}}} \in \mathbb{R}^m$, and the post-treatment outcome vector $\mathbf{y}_{\text{post}} := \mathbf{y}_{\mathcal{T}_2} \in \mathbb{R}^n$.

The evaluation subvector $\mathbf{y}_{\text{eval}}$ is the target pattern to be reconstructed using convex combinations of earlier segments from the pre-treatment period.

SHC by Chen et al. [2024] constructs

$$N = T_0 - m - n + 1$$

overlapping temporal segments of length $m$. Each segment $\widetilde{\mathbf{y}}_{[i,i+m-1]}$ is a contiguous subvector of the smoothed pre-treatment series. These comprise the donor matrix

$$\widetilde{\mathbf{Y}}_{\text{pre}} := \begin{bmatrix} \widetilde{\mathbf{y}}_{[1,m]} & \widetilde{\mathbf{y}}_{[2,m+1]} & \dots & \widetilde{\mathbf{y}}_{[N,N+m-1]} \end{bmatrix} \in \mathbb{R}^{m \times N}.$$

**Example 0.1.** *Suppose we have $T_0 = 60$ months of pre-treatment data and we wish to construct donor segments of length $m = 12$ months to predict a post-treatment horizon of $n = 6$ months. The number of eligible segments is:*

$$N = 60 - 12 - 6 + 1 = 43.$$

*Each donor segment is a vector of 12 consecutive smoothed pre-treatment outcomes.*

In the third step, we solve the following program to obtain the optimal weights:

$$\begin{aligned} \mathbf{w}^* = \underset{\mathbf{w} \in \mathbb{R}^N}{\operatorname{argmin}} \quad & \left\| \widetilde{\mathbf{y}}_{\text{eval}} - \widetilde{\mathbf{Y}}_{\text{pre}} \mathbf{w} \right\|_2^2 + \varsigma \left\| \mathbf{C}_0^\top \mathbf{w} \right\|_2^2 \\ \text{subject to} \quad & \mathbf{w} \geq \mathbf{0}, \quad \mathbf{1}^\top \mathbf{w} = 1. \end{aligned}$$

Let $\mathbf{w}^{\text{SHC}}$ denote the SHC solution, and $\mathbf{w}^{\text{ASHC}}$ the ASHC solution:

$$\mathbf{w}^{\text{ASHC}} = \arg\min_{\mathbf{w} \in \mathbb{R}^N} \left\{ \frac{1}{2\lambda} \|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}} \mathbf{w}\|^2 + \|\mathbf{w} - \mathbf{w}^{\text{SHC}}\|^2 + \varsigma \|\mathbf{C}_0^\top \mathbf{w}\|_2^2 \right\} \quad \text{s.t.} \ \sum_{i=1}^N w_i = 1,$$

where $\lambda > 0$ is the regularization parameter, $\varsigma > 0$ is the penalty parameter for the low-variance directions, and $\mathbf{C}_0 \in \mathbb{R}^{N \times k}$ contains the eigenvectors of the Gram matrix $\mathbf{G} = \widetilde{\mathbf{Y}}^\top \widetilde{\mathbf{Y}}$ corresponding to eigenvalues less than a threshold $\tau > 0$.

## Assumptions

**Assumption 0.1** (Linearity with noise)**.** *There exists* $\mathbf{w}^* \in \mathbb{R}^N$ *such that*

$$\widetilde{\mathbf{y}} = \widetilde{\mathbf{Y}}\mathbf{w}^* + \boldsymbol{\eta},$$

*where* $\boldsymbol{\eta} \in \mathbb{R}^m$ *is a mean-zero noise vector.*

**Assumption 0.2** (Bounded noise)**.** $\|\boldsymbol{\eta}\| \le \delta$ *for some* $\delta > 0$*.*

**Assumption 0.3** (Smoothness of latent trend)**.** *Let* $\ell(t)$ *denote the latent smooth component of the treated potential outcome. Suppose* $\ell(t)$ *is* $H+1$ *times differentiable with* $|\ell^{(H+1)}(t)| < \epsilon$ *for all* $t$ *and some* $H \ge 3$*.*

**Assumption 0.4** (Operator norm bound)**.** $\|\widetilde{\mathbf{Y}}\| \le \gamma$*.*

**Assumption 0.5** (Simplex feasibility)**.** $\mathbf{w}^{SHC} \in \Delta^{N-1} := \{\mathbf{w} \in \mathbb{R}^N : \sum w_i = 1,\ w_i \ge 0\}$*.*

*Proof.* By optimality of $\mathbf{w}^{\text{ASHC}}$ and feasibility of $\mathbf{w}^{\text{SHC}}$:

$$\frac{1}{2\lambda}\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\text{ASHC}}\|^2 + \|\Delta\|^2 + \varsigma\|\mathbf{C}_0^\top(\mathbf{w}^{\text{SHC}} + \Delta)\|_2^2 \le \frac{1}{2\lambda}\epsilon_{\text{SHC}}^2 + \varsigma\|\mathbf{C}_0^\top\mathbf{w}^{\text{SHC}}\|_2^2.$$

□

**Remark 0.1.** *The proof above uses the affine constraint* $\mathbf{1}^\top\mathbf{w} = 1$ *in place of the simplex constraint of SHC. The eigenvalue argument is unaffected by this relaxation because the deviation bound depends only on the strong convexity of the quadratic form defined by*

$$M := \frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{Y}} + 2\mathbf{I} + 2\varsigma\mathbf{C}_0\mathbf{C}_0^\top,$$

*not on the non-negativity of the weights. The affine constraint is enforced through the Lagrange multiplier* $\mu$*, which shifts the right-hand side of the optimality condition but does not alter the conditioning of* $M$*.*

**Lemma 0.1.** *Under (A4), the error from deviating off SHC weights satisfies*

$$\|\widetilde{\mathbf{Y}}\Delta\| \le \gamma\|\Delta\|.$$

**Remark 0.2.** *Lemma 0.1 ensures that deviations of the ASHC weights from the SHC weights, measured in* $\ell_2$ *norm, translate into controlled deviations in the predicted outcomes. The operator norm bound (A4) prevents* $\widetilde{\mathbf{Y}}$ *from amplifying small changes in the weights disproportionately.*

**Lemma 0.2.** *Under (A3), for horizon* $k > 0$*,*

$$|\ell(T_0 + k) - \ell^{SHC}(T_0 + k)| \le b_\epsilon(H, k) := \frac{2\epsilon|k|^{H+1}}{(H+1)!}.$$

*Proof.* Let $\ell(t)$ denote the latent smooth component of the treated outcome. By (A3), $\ell(t)$ is $(H+1)$ times differentiable with $|\ell^{(H+1)}(t)| \le \epsilon$.

**Step 1. Taylor expansion around** $T_0$**.** By Taylor's theorem,

$$\ell(T_0 + k) = \sum_{h=0}^{H} \frac{\ell^{(h)}(T_0)}{h!}\,k^h + R_{H+1}(k),$$

with remainder

$$R_{H+1}(k) = \frac{\ell^{(H+1)}(\xi)}{(H+1)!}\,k^{H+1}, \quad \text{for some } \xi \in [T_0, T_0 + k].$$

**Step 2. Bound the remainder.**

$$|R_{H+1}(k)| \leq \frac{\epsilon}{(H+1)!}\,|k|^{H+1}.$$

**Step 3. SHC approximation of lower-order terms.** By construction, SHC matches the first $H$ terms of this expansion up to error of the same order as the remainder.

**Step 4. Combine bounds.**

$$|\ell(T_0+k) - \ell^{\mathrm{SHC}}(T_0+k)| \leq \frac{2\epsilon}{(H+1)!}\,|k|^{H+1}.$$

□

**Remark 0.3.** *Lemma 0.2 formalizes the SHC bias as a Taylor remainder. The SHC approximation matches derivatives of $\ell(t)$ at $T_0$ up to order $H$, leaving bias decaying like $|k|^{H+1}$. Thus, for smooth latent trends and moderate horizons, SHC bias is negligible.*

# Main Result

**Proposition 0.1.** *Under assumptions (A1)–(A5), the ASHC prediction error is bounded by*

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{ASHC}\| \leq \epsilon_{SHC}\left(1 + \frac{\gamma}{\sqrt{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}}\right) + b_\epsilon(H,k) + \delta.$$

*Proof.* We want to bound the prediction error of the ASHC estimator,

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{ASHC}}\|.$$

**Step 1: Decompose the error using the triangle inequality.** Let $\mathbf{w}^*$ denote the ideal weights from the SHC estimator without the ASHC penalty. Define the difference $\Delta := \mathbf{w}^{\mathrm{ASHC}} - \mathbf{w}^*$. Then,

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{ASHC}}\| = \|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^* - \widetilde{\mathbf{Y}}\Delta\| \leq \|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^*\| + \|\widetilde{\mathbf{Y}}\Delta\|.$$

The first term, $\epsilon_{\mathrm{SHC}} := \|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^*\|$, is the SHC prediction error without augmentation.

**Step 2: Control the second term using operator norm.** By Lemma 0.1, the matrix $\widetilde{\mathbf{Y}}$ satisfies

$$\|\widetilde{\mathbf{Y}}\Delta\| \leq \gamma\|\Delta\|,$$

where $\gamma$ is an operator norm bound capturing how the weights perturbations translate into prediction changes.

**Step 3: Bound the weight difference $\|\Delta\|$.** From Lemma 1 (stated earlier), the ASHC weight vector $\mathbf{w}^{\mathrm{ASHC}}$ stays close to the SHC weights in Euclidean norm:

$$\|\Delta\| = \|\mathbf{w}^{\mathrm{ASHC}} - \mathbf{w}^*\| \leq \frac{\epsilon_{\mathrm{SHC}}}{\sqrt{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}},$$

where the denominator reflects the strength of the ASHC penalty, ensuring stability.

**Step 4: Incorporate smoothness bias and noise.** Finally, the total prediction error also includes the smoothness bias $b_\epsilon(H,k)$ due to approximating the latent function by a truncated Taylor expansion of order $H$, and a noise term bounded by $\delta$ (from assumption (A2)).

Putting all these together, we get

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{ASHC}}\| \leq \epsilon_{\mathrm{SHC}} + \gamma\|\Delta\| + b_\epsilon(H,k) + \delta \leq \epsilon_{\mathrm{SHC}}\left(1 + \frac{\gamma}{\sqrt{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}}\right) + b_\epsilon(H,k) + \delta,$$

which completes the proof. □

## Discussion

Proposition~0.1 decomposes ASHC error into: 1. SHC fit error ($\epsilon_{\text{SHC}}$), 2. amplification from ASHC deviation, 3. smoothness bias $b_\epsilon(H,k)$, 4. noise $\delta$.

The penalty term $\varsigma\|\mathbf{C}_0^\top \mathbf{w}\|_2^2$ reduces deviation but may bias if $\mathbf{w}^*$ has components in $\mathbf{C}_0$ directions. The trade-off between $\lambda$ and $\varsigma$ governs stability vs. fidelity.

## Asymptotic Results

**Theorem 0.1.** *Under (A1)–(A5), as $m \to \infty$ and $H \to \infty$, for fixed $k > 0$,*

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}} w^{ASHC}\| \longrightarrow \delta.$$

*If $\boldsymbol{\eta}$ consists of independent mean-zero sub-Gaussian entries, then*

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}} w^{ASHC}\| \xrightarrow{p} 0.$$

*Proof.* Starting from Proposition 0.1, we have

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}} w^{\text{ASHC}}\| \le \epsilon_{\text{SHC}} \left(1 + \frac{\gamma}{\sqrt{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}}\right) + b_\epsilon(H,k) + \delta.$$

**Step 1. Behavior of $\epsilon_{\text{SHC}}$.** By (A3), $\ell(t)$ is $(H+1)$-times differentiable. Gaussian kernel smoothing in SHC ensures

$$\epsilon_{\text{SHC}} \to 0 \quad \text{as } m \to \infty,\ H \to \infty.$$

Chen et al. [2024]

**Step 2. Behavior of ASHC penalty term.** Since $\epsilon_{\text{SHC}} \to 0$, the inflation factor

$$\frac{\gamma}{\sqrt{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}}\,\epsilon_{\text{SHC}}$$

vanishes asymptotically.

**Step 3. Behavior of smoothness bias $b_\epsilon(H,k)$.** From Taylor's theorem,

$$\ell(T_0 + k) = \sum_{h=0}^{H} \frac{\ell^{(h)}(T_0)}{h!} k^h + R_{H+1}(k),$$

with

$$|R_{H+1}(k)| \le \frac{\epsilon}{(H+1)!}|k|^{H+1}.$$

Thus

$$b_\epsilon(H,k) \le \frac{2\epsilon}{(H+1)!}|k|^{H+1} \to 0 \quad \text{as } H \to \infty.$$

**Step 4. Behavior of noise term.** By (A2), $\|\boldsymbol{\eta}\| \le \delta$. If $\boldsymbol{\eta}$ is i.i.d. sub-Gaussian with variance proxy $\sigma^2$, then

$$\frac{\|\boldsymbol{\eta}\|}{m} \xrightarrow{p} 0.$$

**Step 5. Combine bounds.** Therefore,

$$\limsup_{m\to\infty} \|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}} w^{\text{ASHC}}\| \le \delta.$$

With sub-Gaussian noise, the limit is 0 in probability. ∎

**Corollary 0.1.** *Under (A1)–(A5), the ASHC estimator is asymptotically unbiased up to $\delta$. If $\boldsymbol{\eta}$ is i.i.d. mean-zero sub-Gaussian with variance proxy $\sigma^2$ and bandwidth $h_m \asymp m^{-1/(2H+3)}$, then as $m \to \infty$,*

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}} w^{ASHC}\| = O_p\Big(m^{-H/(2H+3)}\Big).$$

*For large $H$, this approaches the parametric rate $O_p(m^{-1/2})$.*

*Proof.* **Part 1: Consistency.** From Theorem 0.1, the error converges to $\delta$, or 0 in probability under sub-Gaussian noise.

**Part 2: Convergence Rate.** From Proposition 0.1, we have

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}} w^{\mathrm{ASHC}}\| \le \epsilon_{\mathrm{SHC}} \left(1 + \frac{\gamma}{\sqrt{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}}\right) + b_\epsilon(H, k) + \delta.$$

**Step 1. Bias term.** From Lemma 0.2,

$$b_\epsilon(H, k) = O(h_m^H).$$

**Step 2. Stochastic error term.** With sub-Gaussian noise, the smoothed variance is

$$O_p\big((m h_m)^{-1/2}\big).$$

**Step 3. SHC fit error.** Combining,

$$\epsilon_{\mathrm{SHC}} = O_p\big(h_m^H + (m h_m)^{-1/2}\big).$$

**Step 4. Optimal bandwidth.** Equating orders $h_m^H \asymp (m h_m)^{-1/2}$ yields

$$h_m \asymp m^{-1/(2H+3)}.$$

**Step 5. Convergence rate.** Substituting,

$$\epsilon_{\mathrm{SHC}} = O_p\big(m^{-H/(2H+3)}\big).$$

Thus the total error is

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}} w^{\mathrm{ASHC}}\| = O_p\big(m^{-H/(2H+3)}\big).$$

**Step 6. Near-parametric rate.** As $H \to \infty$,

$$\frac{H}{2H+3} \to \tfrac{1}{2}.$$

So the rate approaches $O_p(m^{-1/2})$. ∎

**Remark 0.4.** *The deterministic noise bound $\delta$ in (A2) corresponds to $\|\boldsymbol{\eta}\| \le \delta$. Under sub-Gaussian noise, $\delta = O(\sigma\sqrt{m})$, while kernel smoothing reduces effective noise to $O_p((m h_m)^{-1/2})$. This explains the appearance of the nonparametric rate.*

## References


Yi-Ting Chen, Jui-Chung Yang, and Tzu-Ting Yang. Synthetic historical control for policy evaluation. Available at SSRN: https://ssrn.com/abstract=4995085, September 2024. URL http://dx.doi.org/10.2139/ssrn.4995085.